

Teaching Exam

Subject – Computer Science

**Web Based Application
Development**



Lecture No – 03

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Topics to be Covered



Topic

Go through

HTML

XML.

CSS

Javascript.

HTML

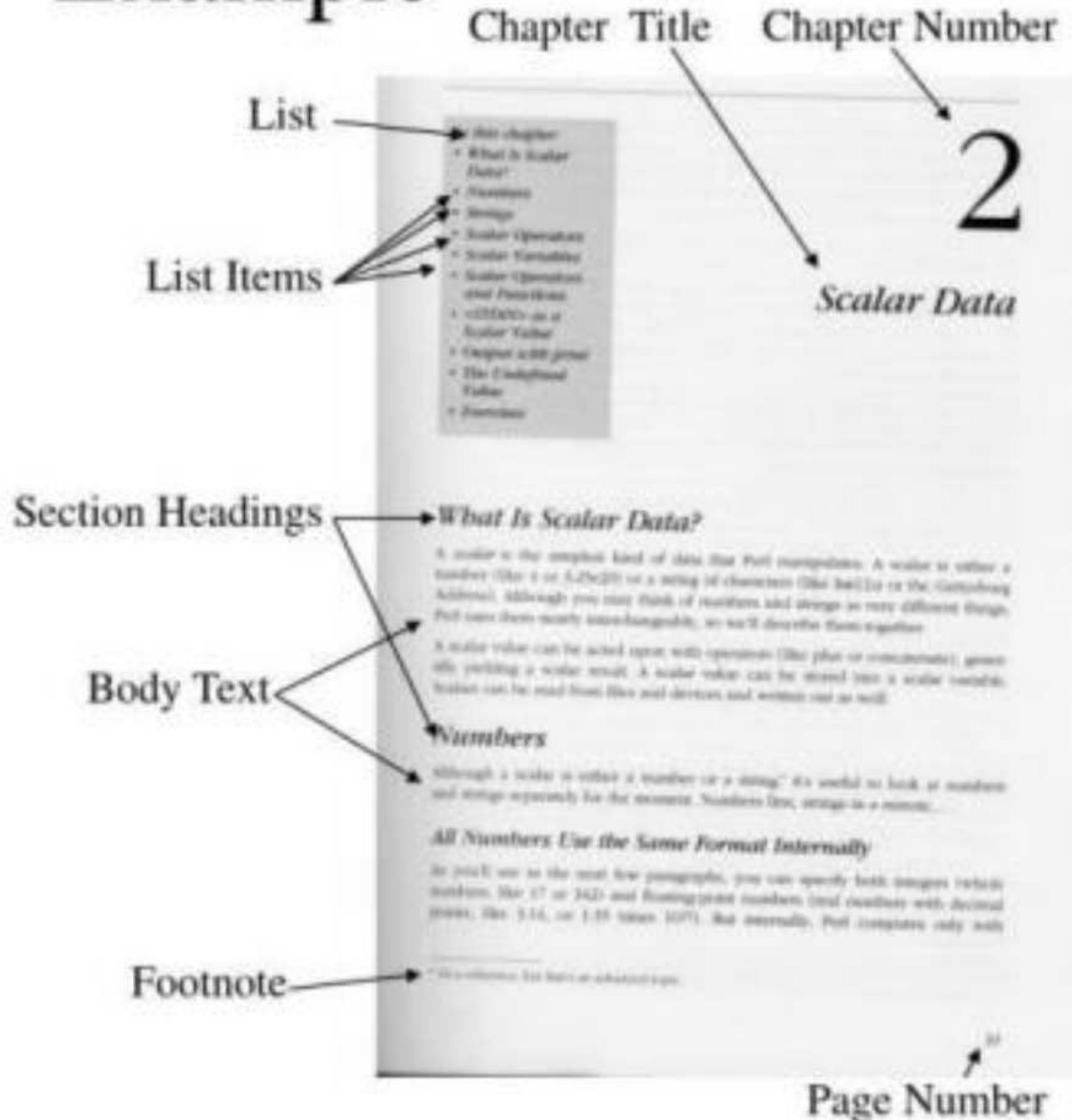
- What does HTML stand for?
 - HyperText Markup Language
 - HTML documents are simply text documents with a specific form.
- What does “markup” mean?
 - Comes from publishing.
 - Documents comprised of content and markup.
 - Content: actual information being conveyed.
 - Markup: side information about the content.

Markup

- Markup information is used to convey the role of different parts of the content and their relationships to each other.

Example

- Here is the first page of chapter two from “Learning Perl”.
- What are the roles of the different pieces of content?



HTML Tags

- Markup is indicated through the use of “tags”.
 - Tags are delimited by angle brackets `<>`
 - Beginning tags
 - Can also include attributes.
 - `<tagname a1=“v1” a2=“v2”>`
 - Ending tags
 - `</tagname>`
 - No attributes.

HTML Document Structure

- Entire document enclosed within `<html>` and `</html>` tags.
- Two subparts parts:
 - Head
 - Enclosed within `<head>` `</head>` tags.
 - Within the head further tags can be used to specify title of page, meta-information, etc.
 - Body
 - Enclosed within `<body>` `</body>` tags.
 - Within the body is the document content to be displayed.

Basic Tag Types

- `<title>`
 - In the head of the document, provides a title for the page.
- `<h1>`, `<h2>`, `<h3>`
 - Different heading levels.
- `<p>`
 - Indicates paragraphs.
- `<br>`
 - Indicates line breaks.

Tag Attributes

- Attributes specified in beginning tag modify and amend tag meaning.
- Each tag has a predefined set of meaningful attributes.
- Unrecognized attributes are ignored.
- Example:
 - Source file for image data.
 -

Anchors and Links

- Key feature of HTML is “linking” between documents.
- Portion of content can be specified as a link using the anchor tag `<a>`
- The href attribute of the anchor tag provides the address of the linked document.
- Example:
 - `<a href=“http://www.cs.unc.edu/~kmp”>`

Anchors

- The target of a link can further specify a specific “anchor” to go to.
 - href=http://server/path/to/page#anchorname
- Defining anchors.
 - Name attribute of an <a> tag.
 -
 - ID attribution of any other tag.
 - <h1 id=“anchorname”>

Embedded Objects

- Other types of data can be embedded into an HTML document.
 - Most commonly: images
- Images indicated using the `<img>` tag.
- The `src` attribute indicates where the image data can be found.
- Example:
 - ``

Function Not Format

- Markup the function of the content without specifying format of the content.
 - Allows end user to decide how function was mapped to appearance.
 - Example: `<em>` for emphasis.
- What are the advantages of doing this?
- What are the disadvantages?

Problems with HTML

- HTML became “polluted” with formatting specific tags and tag attributes.
 - Example: `<b>` for bold.
- Different companies kept adding “custom” tags.
 - Why is this a problem?
- Although human-readable, not always easy for machine to parse and manipulate.
 - What could be hard for the machine but easy for the human?
 - Why might we want a more machine friendly document?

Summary

- HTML as the lingua franca of web documents.
 - The concept of a markup language
 - Annotating document content with role information.
- Motivating XML
 - Extensible framework for structured data.
 - Tree of elements decorated with attributes
 - Separating content, format, and meaning.
- XML details.
 - Rules of well-formedness
 - Well-formed vs. validity.
 - DTD as a mechanism to specify validity.
- XHTML as HTML done right (or at least better).



XML

```
<?xml version="1.0" encoding="UTF-8"?>
<contact>
  <name>John Doe</name>
  <email>john.doe@example.com</email>
  <phone>+1-555-1234</phone>
</contact>
```

with tags we can
have attribute also

Example 2: Product Catalog

XML
SOAP

REST → XML
→ JSON

Graph QL →

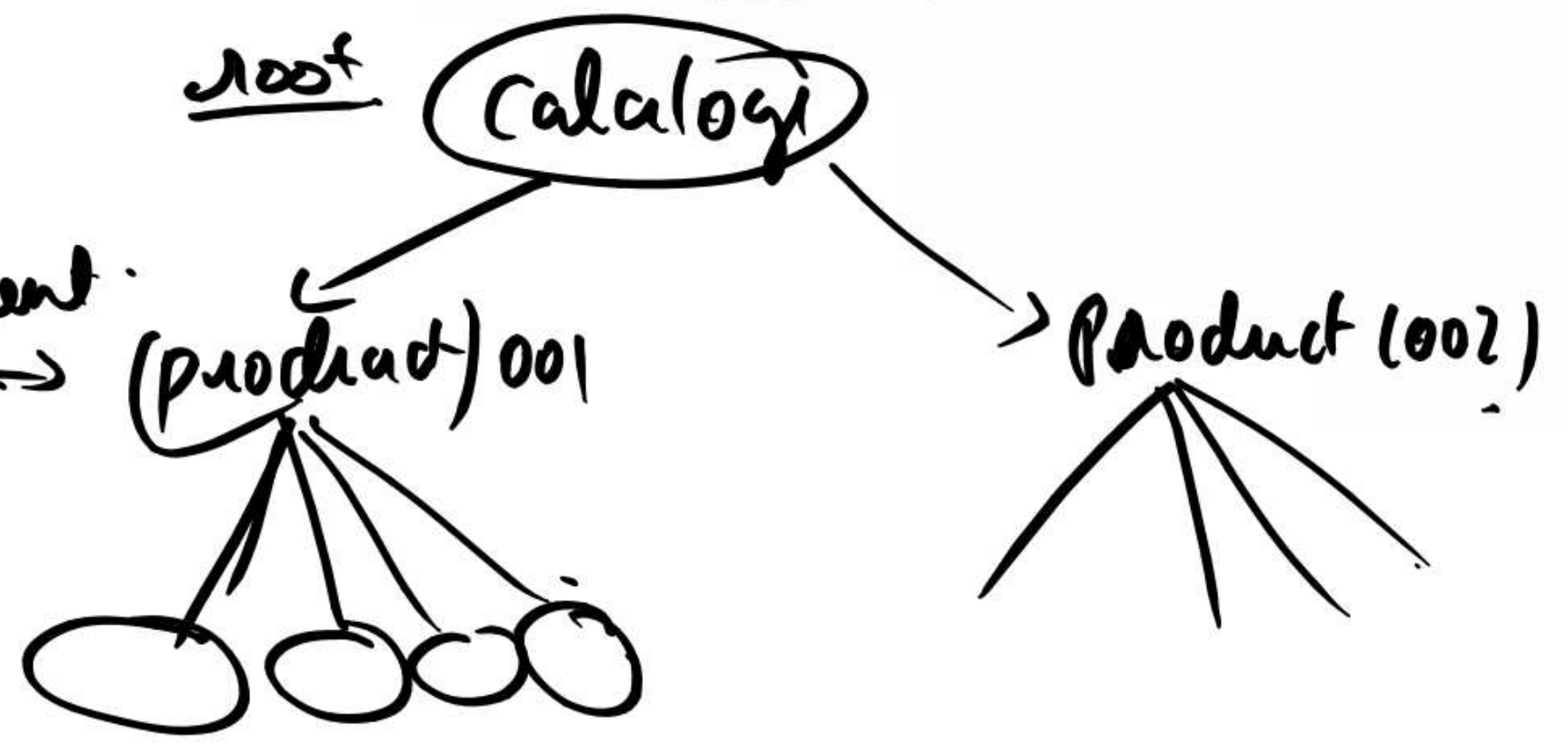
```
<?xml version="1.0" encoding="UTF-8"?>
<catalog>
  <product id="P001">
    <name>Wireless Mouse</name>
    <category>Electronics</category>
    <price currency="USD">25.99</price>
    <stock>120</stock>
  </product>
  <product id="P002">
    <name>Bluetooth Keyboard</name>
    <category>Electronics</category>
    <price currency="USD">45.00</price>
    <stock>75</stock>
  </product>
</catalog>
```

attribute

meta

root

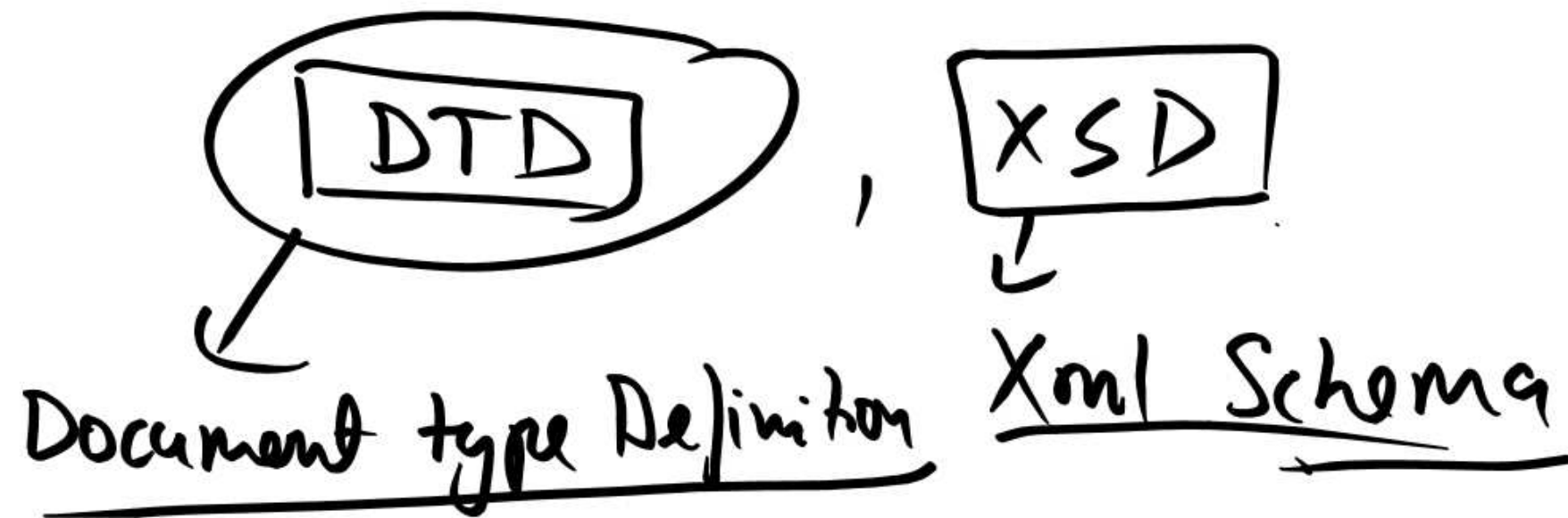
Parent
→



DOM → XML.

Document Object Model (API) - used by Js.

HTML, XML Document - True



<xs: element name = "book" > Root,

<xs: complex = " " > Parent:

<xs: sequence = " " >

DTD

</xs: element . >

JSON.

```
{  //  
  "name": "John Doe",  
  "age": 30,  
  "email": "john.doe@example.com",  
  "isStudent": false  
}
```

Example 2: Product Catalog

```
{
  "products": [
    {
      "id": "P001",
      "name": "Wireless Mouse",
      "price": 25.99,
      "inStock": true
    },
    {
      "id": "P002",
      "name": "Bluetooth Keyboard",
      "price": 45.00,
      "inStock": false
    }
  ]
}
```

 Tips:

- Always wrap property names and string values in **double quotes**.
- Use **braces** `{}` for objects and **square brackets** `[]` for arrays.
- JSON is **less strict than XML** and easier to parse in JavaScript.

✅ DTD stands for Document Type Definition.

Definition:

A DTD defines the *structure* and *rules* for an XML document—what elements and attributes are allowed, and how they can be nested.

You can think of it as:

 A blueprint or grammar for your XML data.

Why do we use DTD?

- ✓ To validate XML files (check if they are "valid" against the defined rules).
- ✓ To ensure consistency across data exchanges.
- ✓ To help parsers understand the expected structure.

Types of DTD

There are two types:

1 Internal DTD //

- Defined **inside** the XML file itself.

2 External DTD =

- Defined in a **separate** file and referenced.

```
<?xml version="1.0"?>
```

```
<!DOCTYPE note [  
  <!ELEMENT note (to, from, heading, body)>  
  <!ELEMENT to (#PCDATA)>  
  <!ELEMENT from (#PCDATA)>  
  <!ELEMENT heading (#PCDATA)>  
  <!ELEMENT body (#PCDATA)>  
>]
```

```
<note>  
  <to>Alice</to>  
  <from>Bob</from>  
  <heading>Reminder</heading>  
  <body>Don't forget the meeting at 10 AM.</body>  
</note>
```

✓ This defines:

- The `<note>` element must contain exactly these four child elements.
- Each child contains **parsed character data** (#PCDATA).

🌐 Were older HTML versions (like HTML 4) able to make web applications?

✅ Yes, they could—but with serious limitations.

HTML itself (any version) is just markup—it defines *structure*, not *behavior*.

So even with HTML 4, people built web apps, but:

HTML 5
2017

- They had to rely heavily on:

- JavaScript (for interactivity)

- Flash, Java Applets, or Silverlight (for rich media and animations)



- The markup was less semantic (lots of `<div>` and `<span>` instead of meaningful tags)
- No native support for:
 - Audio/video embedding without plugins
 - Offline storage
 - Better form controls
 - Responsive design tools

Why was HTML5 a major improvement for web applications?

HTML5 was not just an incremental update—it fundamentally improved how we build rich, interactive apps **without** relying on plugins.

Here's what HTML5 added that older versions **could not do natively**:

Semantic Elements

- `<header>`, `<footer>`, `<article>`, `<nav>` —better structure and accessibility.

Audio and Video

- `<audio>` and `<video>` tags—no Flash needed.

Canvas and SVG

- Drawing graphics dynamically in the browser (`<canvas>`).

✓ Local Storage and Session Storage

- Store data in the browser without cookies.

✓ Form Enhancements

- New input types (`email`, `date`, `range`) and validation.

✓ Offline Web Applications

- Application Cache (now deprecated) and Service Workers.

✅ Geolocation API

- Accessing device location.

✅ WebSockets

- Real-time communication between client and server.

✅ Better Mobile Support

👉 Older HTML versions could create web applications, but:

- They required more hacks, plugins, and workarounds.
- The developer experience was harder.
- Performance and accessibility were worse.

👉 HTML5 made it **much easier, faster, and cleaner** to build modern web apps **natively** in the browser.

1. What is web technology?

a) A collection of tools and techniques used to create and deliver content on the World Wide Web .

b) A type of software that enables users to access and interact with information on the internet

web browser

c) A network of interconnected computers that share information and services

d) A system for storing and retrieving information on the internet

2. Which HTML tag is used to create a hyperlink?

a) <link>

b) <href>

c) <a>

d) <hyperlink>

↓
tag

href
attributed

3. Which of the following is not a web component element?

a) `<shadow>`

b) `<menu>`

c) `<content>`

d) `<element>`

4. Which of the following programming languages is commonly used for server-side scripting in web development?

- a) HTML x]
- b) CSS x ~]
- c) JavaScript] //
- d) PHP

C#

.net 4

5. Which of the following is used to read a HTML page and render it?

a) Web server

b) Web matrix

c) Web browser

d) None of the mentioned

6. Which of the following is a popular front-end framework for building user interfaces in JavaScript?

a) Django

b) Angular

c) Flask

d) Node.js

React

Backend

M.E (A) (N)
M.E (R) (N)

7. What application can one create even before the introduction of HTML5?

- a) Forms ✓
- b) Browser based games ✓
- c) Web applications ✓
- d) Mobile applications ✓ Xaml!

8. Which of the following is not associated with web socket communication?

- a) https
- b) wss
- c) http
- d) ws

9. Why are Images often used on the webpage?

- a) To create graphical buttons or links to other pages ✓
- b) To help the webpage load efficiently ✗
- c) Webpage cannot run/be displayed without at least one image ✗
- d) Because webpage doesn't support pure text ✗

10. What is DOM?

a) Application programming interface ✕

b) Hierarchy of objects in ASP.NET ✕

c) Convention for representing and interacting with objects in html documents ✓

d) None of the mentioned

11. What are tables in web programming?

- a) Display information which are already accessed by the user
- b) Containing more number of rows than columns
- c) To only store data to be accessed later by the user
- ☒ d) Display information in rows and columns used to display all manner of data that fits in a grid

14. What are Forms in web programming?

- a) They are matrices on web page
- b) They are used to create Table data and its information
- c) They allow you to enter data which can be processed to servers
- d) They are the information filled by the server-side

17. Which font format is used in web pages?

- a) EOT
- b) WOFF 2.0
- ☒ c) WOFF
- d) SVG Fonts

24. What is the special feature of modern web applications?

- a) Must load the document to manipulate
- b) Can alter contents without loading document
- c) Can't be altered at all
- d) Remains static

x

26. Which is the storage that allows the caching of web pages and their associated resources?

- a) Offline Web Applications
- b) Cookies ✓✓
- c) Web Databases
- d) FileSystem API

32. React.js is written in which of the following language?

- a) C++
- b) C
- c) Java
- d) JavaScript



2 mins Summary



TEACHING
WALLAH

Topic

. web application & Development

THANK YOU